Diamond Dynamics Project Plan

- Day 1: Data Understanding (March 18, 2026)
  - Handle missing values
  - Convert zero/invalid x,y,z to null/imputed
  - Drop unnecessary columns
  - Apply IQR/Z-Score for outlier detection (carat, price, x,y,z)
- Day 2: EDA & Skewness (March 19, 2026)
  - Perform visual checks for outliers (boxplots)
  - Check skewness of numerical features
  - Apply transformations to skewed variables (price, carat, x,y,z)
  - Generate distribution plots (price, carat, x,y,z)
- Day 3: Categorical EDA & Correlation (March 20, 2026)
  - Create count plots for categorical features (cut, color, clarity)
  - Analyze price variation with carat, cut, color, clarity (boxplots)
  - Generate correlation heatmap for numerical features
  - Create scatterplot matrix (carat, x,y,z, price)
- Day 4: Feature Engineering (March 21, 2026)
  - Generate pairwise relationships (sns.pairplot)
  - Create carat vs. price regression lineplot
  - Create bar plots for avg price per cut, color, clarity
  - Convert price (USD) to INR
  - Derive 'Volume' column (x * y * z)
- Day 5: Adv. Feature Engineering & Selection (March 22, 2026)
  - Derive 'Price per Carat' column
  - Derive 'Dimension Ratio' column
  - Derive 'Carat Category' column
  - Apply feature selection technique
- Day 6: Price Prediction Model Building (March 23, 2026)
  - Perform encoding of categorical columns for regression
  - Split dataset into training and testing sets
  - Build 4-5 ML regression models
  - Build an ANN regression model
- Day 7: Model Evaluation & Segmentation (March 24, 2026)
  - Evaluate all regression models (MAE, MSE, RMSE, R^2)
  - Compare model performances and save best regression model (.pkl)
  - Perform encoding of categorical variables for clustering
  - Implement K-Means clustering
- Day 8: Segmentation Refinement & Streamlit UI (March 25, 2026)
  - Use Elbow Method/Silhouette Score for optimal clusters
  - (Optional) Perform Dimensionality Reduction using PCA
  - Pickle best-performing clustering model (.pkl)
  - Analyze avg price, carat, cut distribution per cluster
  - Assign meaningful names to each cluster
  - Develop Streamlit UI with numeric input fields

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